

Linking Water Quality to Corrosion Behaviour at Steel Sheet Pile Walls: A Machine Learning Study in the Port of Hamburg

Sarah Flohr ¹, Paul Bruck ², Jan Kleinmann ¹, Grit Behrens ¹, and Andreas Kahlfeld²

Abstract: Corrosion of steel sheet pile walls poses a significant challenge for the maintenance of port and waterway infrastructure. This work investigates the relationship between water quality parameters measured at monitoring stations in the Port of Hamburg and site-specific corrosion behaviour, with the aim of identifying environmental drivers of material loss at hydraulic steel structures. A Random Forest model trained on water quality means, structure age, and spatial assignment groups achieves a cross-validated R^2 of 0.719 on an age-adjusted corrosion residual, confirmed by a hold-out R^2 of 0.680. SHAP-based analysis indicates that predictive performance is largely attributable to structure age and spatial group membership. This is consistent with the hydrological data structure: water quality varies between monitoring regions but remains constant within them, which limits the contribution of individual water quality parameters to site-specific corrosion prediction and points to the need for spatially finer-grained input data.

Keywords: Corrosion Prediction, Steel Sheet Pile Walls, Water Quality, Random Forest, Explainable AI

1 Introduction

Steel sheet pile walls are structural elements of port, canal, and coastal infrastructure. They serve to retain river banks, delimit harbour basins and locks, stabilise terrain, and provide flood protection, making them an indispensable component of federal waterway infrastructure. Due to their permanent exposure to water, sediment, and varying chemical conditions, steel sheet pile walls are subject to continuous corrosion [Bu22; HAH07]. The resulting material loss reduces the remaining wall thickness and, over several years, compromises the structural integrity and serviceability of these structures. If the residual wall thickness falls below critical thresholds, the consequences may extend to the stability of adjacent structures; in the event of wall failure, retained soil can escape and contaminants such as oil may leach directly into the waterway, entailing considerable risk of damage and rehabilitation costs.

The study area of this work is the Port of Hamburg. Within the harbour basin, sheet pile walls

1 Hochschule Bielefeld (HSBI), Angewandte Informatik, Artilleriestraße 9, 32427 Minden, Germany, sarah.flohr@hsbi.de,  <https://orcid.org/0009-0009-9901-7841>; jan.kleinmann1@hsbi.de,  <https://orcid.org/0009-0009-2185-5765>; grit.behrens@hsbi.de,  <https://orcid.org/0009-0009-0247-8204>

2 Hochschule Bielefeld (HSBI), Wasser- und Verkehrsbau, Artilleriestraße 9, 32427 Minden, Germany, paul.bruck@hsbi.de,  <https://orcid.org/0009-0005-8963-3182>; andreas.kahlfeld@hsbi.de

are exposed to the varying physical and chemical conditions of a tidally influenced system, where elevated corrosion rates pose a particular challenge for operation and maintenance. Direct corrosion assessment relies on periodic wall thickness measurements, which are both cost- and labour-intensive, resulting in spatially and temporally sparse data that preclude comprehensive condition monitoring. At the same time, continuous water quality time series are available from monitoring stations within the harbour basin, covering parameters such as dissolved oxygen, pH, electrical conductivity, and chloride concentration. This raises the central question of the present work: to what extent can publicly available water quality data help explain site-specific corrosion behaviour of sheet pile walls, and can they support data-driven corrosion assessment? The contribution of this work is a spatial-resolution diagnosis, showing that water quality data capture corrosion behaviour at the group level but not site-specific differences within groups, with direct consequences for monitoring-network design and maintenance planning.

2 Related Work

The corrosion of steel sheet pile walls represents a central challenge for the safety, durability, and economic viability of port and waterway infrastructure. Corrosion in aquatic environments is driven by a combination of physical, chemical, and biological factors (including, for example, dissolved oxygen availability, ionic conductivity, chloride concentration, and pH value [Me06]), interacting in coupled, partly non-linear ways that are difficult to capture through simplified empirical models alone.

The temporal evolution of corrosion loss is commonly described by a power-law model relating cumulative material loss to structure age, widely applied to steel structures under long-term immersion conditions [Ca18; Me03; Me14] and adopted in engineering standards for hydraulic steel structures [Bu22; DI10]. These approaches generally assume stationary boundary conditions and spatially aggregated variables, limiting their capacity to capture local variation and dynamic environmental changes [Ch17]. Climate change further compounds this limitation, as shifting temperature and salinity regimes alter the boundary conditions of corrosion and question the transferability of established empirical values [Fr21; PI22].

A promising alternative lies in exploiting continuously available water quality data. Monitoring stations along federal waterways routinely provide high-resolution time series for relevant parameters, and unlike direct corrosion measurements, which are costly and spatially sparse [HAH07; Me25], these data are available at scale without additional measurement effort.

Machine learning has emerged as a powerful tool for corrosion rate estimation. Yan et al. [Ya20] applied a Random Forest model to marine atmospheric corrosion data, identifying chloride deposition rate, precipitation, and relative humidity as dominant drivers ($R^2 = 0.73$). Yang et al. [Ya24] extended this to depth-varying marine exposure using Gradient Boosting on 856 field samples, finding dissolved oxygen and relative humidity most critical. A broader review by Coelho et al. [Co22] confirms that ensemble methods

consistently outperform classical regression approaches across pipelines, offshore structures, and marine steels [BHZ22; IJM24].

A consistent limitation is the lack of model interpretability. Yang et al. [Ya24] note that certain inputs improve accuracy yet lack physical plausibility, illustrating that predictive performance alone is insufficient for engineering practice. SHapley Additive exPlanations (SHAP), introduced by Lundberg and Lee [LL17], address this by assigning each feature a quantified contribution to individual predictions, enabling both global rankings and instance-level explanations. SHAP has been applied in corrosion contexts [Ya20] and in adjacent environmental domains including water quality assessment and flood susceptibility modelling [Ab26; Sa25].

The present work applies ensemble ML and SHAP-based explainability to corrosion data from steel sheet pile walls in the Port of Hamburg, with the aim of identifying environmental drivers of corrosion and assessing the conditions under which water quality monitoring data can support spatially resolved corrosion prediction.

3 Data and Methodology

The corrosion dataset comprises wall thickness measurements from steel sheet pile walls in the Port of Hamburg. Corrosion loss (in mm) is derived as the difference between the originally installed and the measured residual wall thickness, and related to structure age at the time of measurement (Years Since Construction, t). Figure 1 shows the temporal coverage of available measurements. For the large majority of sites, only a single observation is available; only one site was measured repeatedly. This sparse temporal coverage precludes the reconstruction of individual corrosion trajectories and directly motivates the modelling strategy described below.

Corrosion loss over time is modelled using the power law [Ca18; Me03; Me14]:

$$C(t) = A \cdot t^n, \quad (1)$$

where $C(t)$ is the cumulative corrosion loss in mm, A the initial corrosion rate, and n a dimensionless exponent describing the deceleration of corrosion over time. A global fit over the full dataset yields $C(t) = 0.0800 \cdot t^{0.7135}$, which serves as the reference curve for all subsequent analyses. Prior to fitting, measurement positions within each site are averaged to avoid treating repeated measurements at the same structure as statistically independent observations.

Since the majority of sites provide only a single observation, a station-level trajectory fit is not feasible. Instead, an age-adjusted residual C_{residual} is introduced as the target variable, defined as the deviation of the observed corrosion loss from the age-dependent expectation of the global fit:

$$C_{\text{residual},i} = C_{\text{obs},i} - \hat{C}(t_i), \quad (2)$$

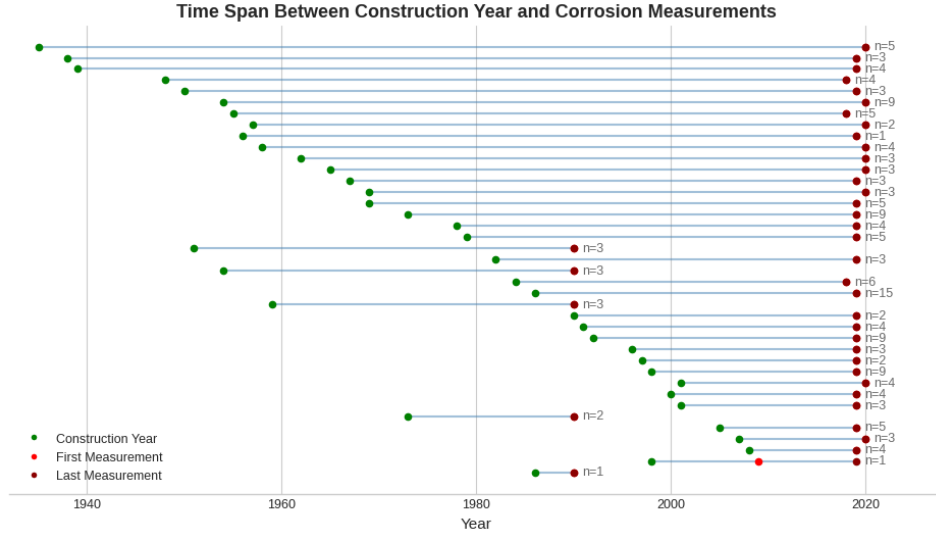


Fig. 1: Temporal coverage of corrosion measurements per site. Green markers indicate the year of construction, orange markers the first measurement, and red markers the last available measurement. The annotation gives the number of sheet piles per site for which both a construction year and at least one measurement are available. The majority of sites have only a single measurement point.

where $C_{obs,i}$ is the observed corrosion loss, t_i the age of site i at the time of measurement, and $\hat{C}(t_i) = A_{fit} \cdot t_i^{n_{fit}}$ the global power law evaluated at age t_i . Positive values indicate corrosion in excess of the global expectation; negative values indicate below-average corrosion. Because the subtracted term $\hat{C}(t_i)$ varies with t_i , $C_{residual}$ is station-individual and is largely decoupled from the global age trend.

Water quality time series from monitoring stations within the harbour basin are used to characterise local chemical and physical exposure conditions. The dataset covers water temperature, dissolved oxygen, oxygen saturation, chloride concentration, dissolved organic carbon (DOC), pH, and electrical conductivity.

Figure 2 shows the temporal coverage and measurement frequency for each parameter.

A central methodological challenge is the spatial assignment between water quality monitoring stations and corrosion measurement sites. As monitoring stations are distributed at a coarser spatial resolution than the corrosion sites, a one-to-one assignment is not feasible. Instead, the assignment is performed via structural groups, each comprising multiple corrosion parents that are jointly assigned to a single water quality station. These groups were defined by a hydraulic engineering expert based on flow behaviour and spatial proximity, and represent hydrologically homogeneous zones. As a consequence, all corrosion sites within the same group receive identical water quality values. The dataset was further restricted to sites for which all seven water quality parameters were actually recorded at the assigned

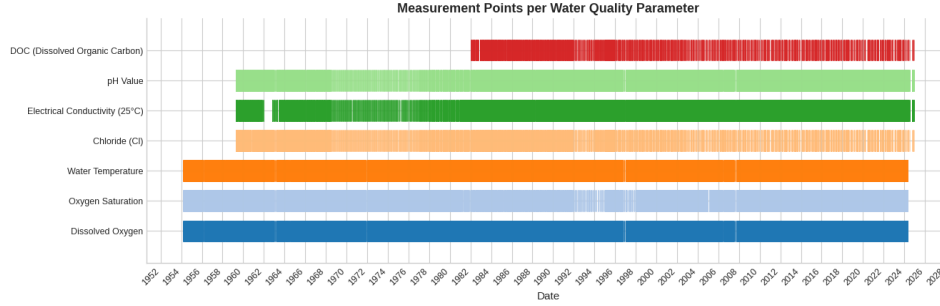


Fig. 2: Temporal coverage and measurement frequency of the seven water quality parameters at the monitoring stations in the Port of Hamburg. Regular measurements are available for most parameters from the 1950s onwards, substantially exceeding the temporal density of the corrosion dataset.

monitoring station, yielding five monitoring stations assigned to 39 corrosion parents and 153 individual measurement sites.

The feature matrix for each corrosion site comprises the temporal mean of each of the seven water quality parameters over the full available measurement period of the assigned monitoring station, the structure age at the last measurement (Years Since Construction), and the assignment group encoded as a one-hot variable. Since most sites provide only a single observation, the structure age corresponds to the age at that observation. Temporal averaging of water quality data represents a first exploratory approximation, assuming that long-term mean values adequately represent the cumulative chemical exposure over the structure’s lifetime.

The assignment group is deliberately retained as a feature despite its close relationship with the water quality values. This enables the contribution of the group structure to be made explicit in the model and quantified through ablation experiments comparing the full model against a model excluding the group variable. The target variable throughout is C_{residual} as defined in Equation 2.

A Random Forest regression model is trained on C_{residual} to identify relevant drivers of corrosion behaviour. The model is evaluated using 5-fold cross-validation, supplemented by hold-out validation to assess generalisation. Feature contributions are assessed using permutation importance, computed on the hold-out set to avoid overfitting bias, and complemented by a SHAP-based analysis to examine contributions at the level of individual sites. A binary classification model is additionally trained, with the target variable indicating whether $C_{\text{residual}} > 0$, i.e. whether a site corrodes more than globally expected.

4 Preliminary Results

A Random Forest regression model trained on all features achieves a cross-validated R^2 of 0.719 ± 0.087 on C_{residual} , substantially more stable than a model using water quality features alone ($R^2 = 0.388 \pm 0.083$). The cross-validation estimate is confirmed by a hold-out

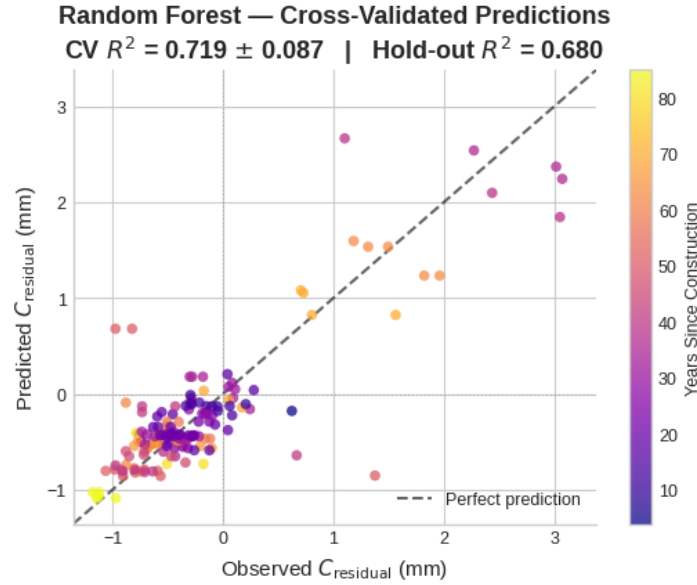


Fig. 3: Cross-validated predictions of the Random Forest regression model against observed C_{residual} values (CV $R^2 = 0.719 \pm 0.087$, hold-out $R^2 = 0.680$). The dashed line indicates perfect agreement. Colour encodes structure age; sites with high positive residuals are predominantly older. Predictions are well-calibrated in the central range but systematically underestimate strongly corroding sites.

R^2 of 0.680. Figure 3 shows the cross-validated predictions against observed C_{residual} values. Predictions are well-calibrated in the central range; sites with strongly positive residuals, indicating corrosion substantially above the global expectation, are systematically underestimated, consistent with their underrepresentation in the training data.

Permutation importance on the hold-out set attributes predictive performance predominantly to three features: structure age (Years Since Construction, 0.475), mean dissolved oxygen (0.204), and a single assignment group (wq_group_3, 0.157), while all remaining water quality features contribute negligibly. The moderate negative association between structure age and C_{residual} suggests that the global fit tends to overestimate corrosion loss at older structures.

A direct comparison between the full model ($R^2 = 0.719 \pm 0.087$) and a model excluding the explicit group variable ($R^2 = 0.718 \pm 0.087$) reveals virtually identical performance. This reflects the hydrological consistency of the spatial assignment: the groups were defined by a hydraulic engineering expert based on flow behaviour and spatial proximity, and represent hydrologically homogeneous zones. Within each zone, water quality values are by construction constant, so the water quality means already implicitly encode the group structure. The practical consequence is that site-specific corrosion differences within a hydrologically homogeneous zone cannot be fully explained by water quality data alone.

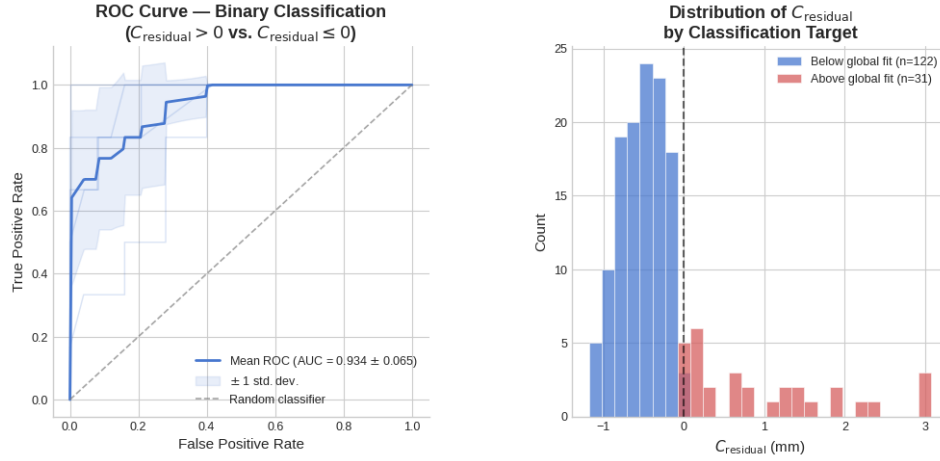


Fig. 4: Left: ROC curve of the binary classification model ($C_{\text{residual}} > 0$ vs. $C_{\text{residual}} \leq 0$) across five cross-validation folds ($\text{AUC} = 0.934 \pm 0.065$). Right: distribution of C_{residual} by class ($n = 122$ below vs. $n = 31$ above the global fit).

As a preliminary robustness check, the model was additionally evaluated under leave-one-group-out cross-validation, holding out one of the five monitoring groups per fold. The pooled out-of-group R^2 falls below the global-mean baseline, indicating that the random cross-validated performance partly reflects group identity shared between training and test sites. A larger dataset with more monitoring groups would be required to quantify generalisation to unseen locations reliably.

A binary classification model predicting whether $C_{\text{residual}} > 0$ achieves a cross-validated ROC-AUC of 0.934 ± 0.065 , indicating that sites corroding above the global expectation are reliably distinguished from those below. Figure 4 shows the ROC curve and the distribution of C_{residual} by class. Of the 153 sites, 122 lie below and 31 above the global fit. The two classes show little overlap in the residual space, and the pronounced class imbalance should be considered when interpreting the AUC, as structure age and group membership contribute substantially to discriminative performance.

SHAP values were computed for the Random Forest model to assess feature contributions at the site level. Figure 5 shows the beeswarm plot excluding group variables. Structure age dominates with a wide, non-monotone SHAP spread (-0.75 to $+1.10$). Among water quality parameters, only mean dissolved oxygen (DO) shows a noteworthy contribution (-0.25 to $+0.60$), with low DO values tending to be associated with elevated C_{residual} , consistent with the role of oxygen availability in aquatic corrosion processes. All remaining parameters contribute negligibly.

Closer inspection confirms that structure age acts as the primary differentiating feature within hydrologically homogeneous monitoring regions. Sites sharing the same parent and age receive identical SHAP values, indicating that age captures location identity rather

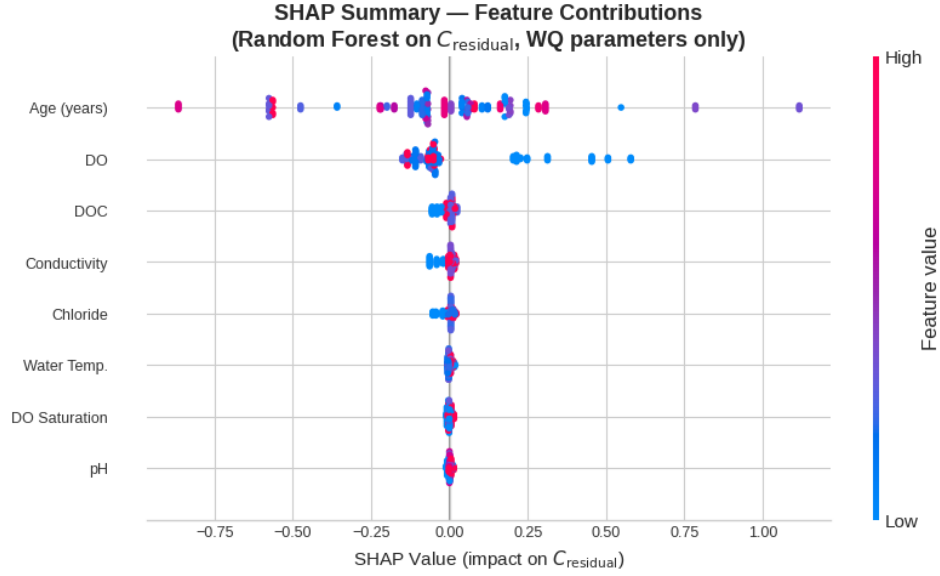


Fig. 5: SHAP beeswarm plot for the Random Forest model on C_{residual} (group variables excluded). Each point represents one site; colour encodes the feature value (red: high, blue: low). Structure age dominates with a wide, non-monotone SHAP spread. Among water quality parameters, only mean dissolved oxygen (DO) shows notable variation; low DO values are associated with elevated C_{residual} . All other parameters contribute negligibly.

than a genuine ageing effect. Within the same monitoring region, sites of different ages exhibit opposing SHAP signs, confirming that structure age captures location-specific differences rather than a direct ageing effect, as it remains the only feature able to account for within-group variation in C_{residual} where water quality is constant.

5 Discussion

The Random Forest model achieves a cross-validated R^2 of 0.719 on the age-adjusted target variable C_{residual} , confirmed by a hold-out R^2 of 0.680, indicating stable generalisation. Since test sites were selected randomly during cross-validation, the reported R^2 likely overestimates generalisation to entirely unseen locations. A preliminary leave-one-group-out check supports this, though a larger dataset is needed for a robust assessment.

The near-identical performance of the full model ($R^2 = 0.719$) and a model excluding the explicit group variable ($R^2 = 0.718$) reflects the hydrological consistency of the spatial assignment: water quality means implicitly encode group structure, so site-specific differences within a group cannot be explained by water quality data alone.

The SHAP analysis confirms this. Structure age dominates predictions not through a genuine

ageing effect but as the only feature differentiating sites within a group. Among water quality parameters, only mean dissolved oxygen shows a noteworthy association with C_{residual} , with lower values tending to correspond to higher corrosion residuals, consistent with the role of oxygen in aquatic corrosion. Whether this reflects a genuine chemical signal cannot be resolved from the available data. The spatial resolution of the monitoring network is too coarse to disentangle chemical signal from location effects, and 99.3 % of sites provide only a single measurement, precluding individual trajectory reconstruction.

Conclusion and Outlook Water quality data reflect corrosion behaviour at the level of hydrologically homogeneous monitoring groups, providing a consistent basis for group-level assessment of groups represented in the training data. A preliminary leave-one-group-out check suggests that prediction at unobserved locations is substantially harder. For site-specific prediction within groups, water quality data alone are insufficient, as finer-grained factors such as local flow conditions or structural properties govern the remaining variability. Future work should identify spatially finer-grained predictors, extend the feature set towards richer temporal statistics, and expand the corrosion dataset to multiple measurements per site. A variance-partitioning analysis over structure age, group membership, and water quality would further quantify how much predictive signal originates from the spatial group structure rather than the water quality parameters. The long-term objective is reliable corrosion risk assessment at sites with sparse inspection records, supporting more efficient maintenance planning along federal waterways.

References

- [Ab26] Ab Karim, M. A.; Wan Ismail, W. Z.; Mohd Shuib, F. M.; Ab Aziz, N. A.; Ghazali, A. K.: Water Quality Monitoring and Assessment Using Machine Learning: A Review of Formulation, Modeling Approaches, and Explainable Artificial Intelligence. *Environments* 13 (5), 2026, doi: 10.3390/environments13050267.
- [BHZ22] Ben Seghier, M. E. A.; Höche, D.; Zheludkevich, M.: Prediction of the Internal Corrosion Rate for Oil and Gas Pipeline: Implementation of Ensemble Learning Techniques. *Journal of Natural Gas Science and Engineering* 99, 2022, doi: 10.1016/j.jngse.2022.104425.
- [Bu22] Bundesanstalt für Wasserbau: BAW-Empfehlung: Spundwanddickenmessungen in Häfen und an Wasserstraßen: Grundlagen, Planung, Durchführung, Auswertung und Interpretation (ESM), BAW-Merkblatt, -Empfehlung und -Richtlinie, Karlsruhe: Bundesanstalt für Wasserbau (BAW), 2022, URL: https://izw.baw.de/publikationen/empfehlungen/0/BAWEmpfehlung_Spundwanddickenmessung_ESM_2022.pdf.
- [Ca18] Cai, Y.; Zhao, Y.; Ma, X.; Zhou, K.; Chen, Y.: Influence of Environmental Factors on Atmospheric Corrosion in Dynamic Environment. *Corrosion Science* 137, pp. 163–175, 2018, doi: 10.1016/j.corsci.2018.03.042.
- [Ch17] Chico, B.; de la Fuente, D.; Díaz, I.; Simancas, J.; Morcillo, M.: Annual Atmospheric Corrosion of Carbon Steel Worldwide: An Integration of ISOCORRAG, ICP/UNECE and MICAT Databases. *Materials* 10 (6), 2017, doi: 10.3390/ma10060601.
- [Co22] Coelho, L. B.; Zhang, D.; Van Ingelgem, Y.; Steckelmacher, D.; Nowé, A.; Terryn, H.: Reviewing Machine Learning of Corrosion Prediction in a Data-Oriented Perspective. *npj Materials Degradation* 6 (1), p. 8, 2022, doi: 10.1038/s41529-022-00218-4.

- [DI10] DIN: DIN EN 1993-5:2010-12: Eurocode 3: Bemessung und Konstruktion von Stahlbauten – Teil 5: Pfähle und Spundwände; Deutsche Fassung EN 1993-5:2007 + AC:2009, 2010, doi: <https://dx.doi.org/10.31030/1723936>.
- [Fr21] Fritsch, U. et al.: Klimawirkungs- und Risikoanalyse für Deutschland 2021: Teilbericht 3: Klimarisiken im Cluster Wasser, Climate Change 22/2021, Umweltbundesamt, 2021.
- [HAH07] Houyoux, C.; Alberts, D.; Heeling, A.: Design Method for Steel Structures in Marine Environment Including the Corrosion Behaviour, Directorate-General for Research and Innovation, European Commission, 2007, URL: <https://op.europa.eu/publication-detail/-/publication/d742313f-f26f-4171-8f05-e5f78f2e4c6f>.
- [IJM24] Imran, M. M. H.; Jamaludin, S.; Mohamad Ayob, A. F.: A Critical Review of Machine Learning Algorithms in Maritime, Offshore, and Oil & Gas Corrosion Research: A Comprehensive Analysis of ANN and RF Models. Ocean Engineering 295, p. 116796, 2024, doi: [10.1016/j.oceaneng.2024.116796](https://doi.org/10.1016/j.oceaneng.2024.116796).
- [LL17] Lundberg, S. M.; Lee, S.-I.: A Unified Approach to Interpreting Model Predictions. In: Advances in Neural Information Processing Systems. Vol. 30, pp. 4765–4774, 2017, doi: <https://doi.org/10.48550/arXiv.1705.07874>.
- [Me03] Melchers, R. E.: Mathematical modelling of the diffusion controlled phase in marine immersion corrosion of mild steel. Corrosion Science 45 (5), pp. 923–940, 2003, doi: [https://doi.org/10.1016/S0010-938X\(02\)00208-1](https://doi.org/10.1016/S0010-938X(02)00208-1).
- [Me06] Melchers, R. E.: Modelling Immersion Corrosion of Structural Steels in Natural Fresh and Brackish Waters. Corrosion Science 48 (12), pp. 4174–4201, 2006, doi: [10.1016/j.corsci.2006.04.012](https://doi.org/10.1016/j.corsci.2006.04.012).
- [Me14] Melchers, R. E.: Modelling Long Term Corrosion of Steel Infrastructure in Natural Marine Environments. In (Melchers, R. E.; Liengen, T.; Féron, D.; Basséguy, R.; Beech, I. B., eds.): Understanding Biocorrosion. Vol. 66, Woodhead Publishing Series in Materials, Woodhead Publishing, Amsterdam, pp. 213–241, 2014.
- [Me25] Melchers, R. E.; Jeffrey, R.; Chaves, I. A.; Petersen, R. B.: Predicting Corrosion for Life Estimation of Ocean and Coastal Steel Infrastructure. Materials and Corrosion 76 (6), pp. 776–789, 2025, doi: [10.1002/maco.202314201](https://doi.org/10.1002/maco.202314201).
- [PI22] PIANC: Managing Climate Change Uncertainties in Selecting, Designing and Evaluating Options for Resilient Navigation Infrastructure: Technical Note No. 1, Permanent Task Group for Climate Change, 2022, URL: <https://pianc.org.au/wp-content/uploads/2022/04/PTGCC-Technical-Note-1-Managing-Climate-Change-Uncertainties-April-2022.pdf>, visited on: 02/24/2026.
- [Sa25] Salih, M. et al.: A Perspective on Explainable Artificial Intelligence Methods: SHAP and LIME. Advanced Intelligent Systems 7 (1), p. 2400304, 2025, doi: [10.1002/aisy.202400304](https://doi.org/10.1002/aisy.202400304).
- [Ya20] Yan, L.; Diao, Y.; Lang, Z.; Gao, K.: Corrosion Rate Prediction and Influencing Factors Evaluation of Low-Alloy Steels in Marine Atmosphere Using Machine Learning Approach. Science and Technology of Advanced Materials 21 (1), pp. 359–370, 2020, doi: [10.1080/14686996.2020.1746196](https://doi.org/10.1080/14686996.2020.1746196).
- [Ya24] Yang, J.; Zou, D.; Zhang, M.; Que, Z.; Liu, T.; Zhou, A.; Li, Y.: Marine Steel Corrosion Prediction and Zonation Using Feature Extraction and Machine Learning in the Seas Around China. Ocean Engineering 314, p. 119649, 2024, doi: [10.1016/j.oceaneng.2024.119649](https://doi.org/10.1016/j.oceaneng.2024.119649).